



Contamination of ready-to-eat street food in Pakistan with *Salmonella* spp.: Implications for consumers and food safety

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ARTICLE INFO

Article history:

Received 30 December 2020

Received in revised form 13 March 2021

Accepted 18 March 2021

Keywords:

RTE food

Microbial contamination

Salmonellosis

Antibiotic sensitivity

ABSTRACT

Objectives: Ready-to-eat (RTE) food sold in Quetta, Pakistan was assessed for microbial contamination. **Methods:** Equal numbers of samples were collected from four categories of RTE food – burgers, shawarma, pizza and sandwiches – from January 2018 to December 2018. Microbial contamination of individual food samples was assessed by quantifying the total aerobic count obtained from plating samples on bacterial growth medium. *Salmonella* spp. serovars were identified using polymerase chain reaction.

Results: Approximately 38% (121/320) of RTE food samples were not fit for human consumption. The most contaminated type of RTE food was shawarma (49%). Microbial contamination of food samples was higher in summer compared with the other seasons. Approximately 40% (49/121) of food samples that were not fit for human consumption were contaminated with *Salmonella* spp. *Salmonella enteritidis* (69%) and *Salmonella typhimurium* (31%) were the only serovars among the samples testing positive for *Salmonella* spp. Of the 49 samples with high microbial counts, *S. enteritidis* was present in 34 samples and *S. typhimurium* was present in 15 samples. The antibiotic sensitivity results demonstrated that both *S. enteritidis* and *S. typhimurium* were resistant to amoxicillin. In addition, *S. enteritidis* was resistant to chloramphenicol and erythromycin, and *S. typhimurium* presented high resistance to erythromycin. Both *S. typhimurium* and *S. enteritidis* were highly sensitive to kanamycin.

Conclusion: RTE food sold by street vendors in Quetta was found to be contaminated with *Salmonella* spp. and poses a great health risk to consumers. As such, consumption should be avoided, and the health authorities should take stringent action to ensure the quality of street food in order to reduce the healthcare burden.

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Introduction

Developing countries such as Pakistan still suffer with foodborne illnesses, and salmonellosis is the most common (Akhtar et al., 2014; Hull-Jackson et al., 2019; Siddiqui et al., 2015). Salmonellosis is mainly caused by *Salmonella enterica* (Van Cauteren et al., 2017). Several other bacterial pathogens such as *Escherichia coli*, *Listeria monocytogenes*, *Bacillus cereus* and

Staphylococcus aureus are frequently involved in foodborne illnesses (Olaimat et al., 2020; Van Cauteren et al., 2017). Over recent decades, the consumption of fast food/ready-to-eat (RTE) food has increased significantly (Saghaian and Mohammadi, 2018; Xue et al., 2016). RTE food is normally prepared and cooked on streets in developing countries, and are consumed by millions of people each day. Street food is popular because they can be prepared and served quickly at cheaper prices than in sit-down restaurants, and they are available on practically every street (Ashakiran and Deepthi, 2012; Magnini et al., 2013). Pizza, shawarma, sandwiches, burgers and French fries are among the favourite fast foods in developing countries such as Pakistan. Street food is also an important attraction for tourists (Henderson et al.,

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2012). Unfortunately, in developing countries, street food is mainly prepared and sold on streets without taking proper hygienic measures (Teffo, 2017), and consumers are at risk of bacterial contamination which can lead to gastroenteritis and other foodborne illnesses (Ahmed et al., 2017; Alimi, 2016; Atter et al., 2015; De Lima et al., 2019; Oscar, 2020). The main sources of bacterial contamination are dirty utensils, impure and substandard cooking oil, and unhygienic practices by vendors (Asiegbu et al., 2016). Often, running tap water is not available at the foot carts, and utensils are washed in a bucket filled with water (Asiegbu et al., 2016). Another factor which is often neglected in developing countries is that food carts on streets are open to the environment, without any precautions to keep the food safe from contamination by humans or flies, and food is often kept at high ambient temperatures that favour bacterial growth. Numerous relevant factors influence the microbial contamination of food, and this food may be hazardous to public health (Chlebicz and Śliżewska, 2018; Niyonzima et al., 2015).

Salmonellosis is an infection caused by *Salmonella* spp., and humans often become infected after consumption of contaminated food and water. Foodborne illnesses have been reported frequently from developing countries, such as Thailand, China and India (Ananchaipattana et al., 2012; Chen et al., 2014), and foodborne salmonellosis may represent up to one-third of all foodborne illnesses (Ananchaipattana et al., 2012; Chen et al., 2014; Manguit and Fang, 2013). In fact, due to under-reporting, salmonellosis case numbers are likely to be much higher in developing countries than the numbers stated in official reports (Ahmed et al., 2017; Chlebicz and Śliżewska, 2018).

Several foodborne pathogens such as *Salmonella* spp., *Campylobacter* spp., *Listeria* spp. and *Yersinia* spp. can spoil food (Chlebicz and Śliżewska, 2018), and ideally all served food should be free of all microbial pathogens. To the authors' knowledge, this is the first study to investigate the microbiological quality of RTE food sold in Quetta, Pakistan.

Methods

Sample collection

In total, 320 RTE food samples (burgers, shawarma, pizza and sandwiches) were purchased from street food vendors, selected at random. Sampling was performed in all four seasons between January 2018 and December 2018. Samples were collected in sealed sterile bags, and a cold chain was maintained during transportation. Samples were processed on the same day.

Sample processing and total aerobic count

Food samples (25 g) were homogenized by vigorous shaking in buffered peptone water. To determine the total bacterial count, samples were further diluted to obtain countable colonies on plate count agar. Briefly, plate count agar was streaked with 1-mL aliquots of diluted (1:10) samples and incubated at 37 °C for 48 h. After incubation, the bacterial colonies on agar plates were

counted and multiplied by the dilution factor to achieve the total bacterial count.

Isolation and identification of *Salmonella* spp

For selective enrichment of *Salmonella* spp., 1-mL aliquots from the diluted homogenized samples were mixed in Mueller Kauffman Tetrathionate Novobiocin broth and Rappaport–Vassiliadis broth, and incubated for 24 h at 37 °C and 41.5 °C, respectively. The selective enriched colonies were further transferred to xylose-lysine deoxy-cholate and salmonella shigella agar. The typical black colonies were further subjected to Gram staining and an array of biochemical tests such as Simmons citrate, oxidase, indole production, lysine decarboxylase, catalase, urease, H₂S and mannitol fermentation for further confirmation.

~~Conventional and multiplex PCR based identification of *Salmonella* spp~~

~~DNA was extracted from isolated *Salmonella* spp. colonies by the heat lysis method. Briefly, a single colony was picked, dissolved in polymerase chain reaction (PCR) grade water and boiled at 100 °C for 15 min. After boiling, samples were centrifuged. The supernatant was used instantly as a DNA template or stored at –20 °C for later use. For identification to genus level, *invA* was targeted and amplified with a specific set of primers, as described by Oliveira et al. (2003). For identification to species level, a specific set of primers was employed to amplify *fliC* (*S. typhimurium*), *sefA* (*S. enteritidis*), *fliC* (*S. typhi*), *fliCa* (*S. paratyphi A*) and *SPAB_01124* (*S. paratyphi B*), as described previously (Zishiri et al., 2016). The primers used in PCR and PCR reagents were purchased from Macrogen (Seoul, South Korea).~~

Antimicrobial sensitivity

Antimicrobial sensitivity was performed using the Kirby–Bauer disc diffusion method (Garedew et al., 2015). To check the sensitivity of drugs, 11 antibiotic-impregnated discs were placed on the agar in two groups: amikacin, gentamicin, ampicillin and streptomycin (10 µg each); and amoxicillin, ceftriaxone, chloramphenicol, kanamycin, ciprofloxacin, nitrofurantoin, nalidixic acid (30 µg each). The impregnated discs were placed on the surface of culture plates and incubated at 30 °C for 24 h. The diameters of the zones of inhibition were recorded in millimetres. Classification was established as sensitive, intermediate or resistant in accordance with the interpretive chart.

Results

Contamination of RTE food in different seasons

All RTE food samples ($n = 320$) were collected and analysed during the four seasons of the year, and all were found to be contaminated with at least one microbial pathogen. Of these, 62.18% ($n = 199$) of the contaminated samples had an 'acceptable'

Table 1
Microbial contamination of ready-to-eat (RTE) food items during different seasons.

Food type	Unacceptable for consumption				
	Winter ($n = 20$)	Spring ($n = 20$)	Summer ($n = 20$)	Autumn ($n = 20$)	Total
Pizza ($n = 80$)	4	5	10	4	23 (28.75%)
Burger ($n = 80$)	5	6	12	5	28 (35%)
Shawarma ($n = 80$)	6	7	18	8	39 (48.75%)
Sandwich ($n = 80$)	5	5	14	7	31 (38.75%)
Total contaminated samples	20	23	54	24	121 (37.81%)

level of microbial contamination and were fit for consumption. For the remaining 121 contaminated samples (37.81%), the aerobic plate count (APC) was $>5 \log_{10}$ colony-forming units (cfu)/g, and these samples were therefore classed as 'unacceptable for consumption' (Table 1). Among the samples analysed, the shawarma samples were the most contaminated (48.7%), followed by sandwiches, burgers and pizza. The highest level of contamination occurred in the summer (Table 1).

Quantification of total aerobic count

Food with $>5 \log_{10}$ cfu/g microbes is considered to be fit for consumption. To differentiate between food items that were and were not acceptable for human consumption, the authors were interested to quantify the intensity of microbial contamination. The results demonstrated that the mean APC level in samples ($n = 121$) with $>5 \log_{10}$ cfu/g ranged from $6 \log_{10}$ cfu/g to $9.8 \log_{10}$ cfu/g over the four seasons, and the highest level of contamination was observed in the summer (Figure 1).

Identification and quantification of *Salmonella* spp. in RTE food samples

The contaminated samples ($n = 121$) were further investigated for the prevalence of *Salmonella* spp. These are considered to be the most potent pathogens for foodborne illnesses. Multiplex PCR was used as well as conventional biochemical tests (results not shown) (Figure 2). Successful amplification of *S. enteritidis* and *S. typhimurium* was visualized, and no samples were found to be contaminated with *S. typhi* or *S. paratyphi* A or B. The representative results of multiplex PCR are shown in Figure 2.

The multiplex PCR results showed that 49 of 121 RTE food samples (40.4%) were contaminated with *Salmonella* spp., of which 34 (69.39%) were identified as *S. enteritidis* and 15 (30.61%) were identified as *S. typhimurium* (Figure 2).

Antibiotic sensitivity profile of isolated *Salmonella* spp

The antibiotic susceptibility pattern of *Salmonella* spp. isolated from RTE food is presented in Table 2. All *S. enteritidis* isolates (34/34) were resistant to erythromycin and chloramphenicol, and all *S. typhimurium* isolates were resistant to erythromycin. *S. typhimurium* and *S. enteritidis* also showed a high level of resistance to amoxicillin. All *S. typhimurium* isolates were sensitive to kanamycin, and *S. enteritidis* also showed high sensitivity (25/34, 73.53%) as only one isolate was resistant to this drug (Table 2).

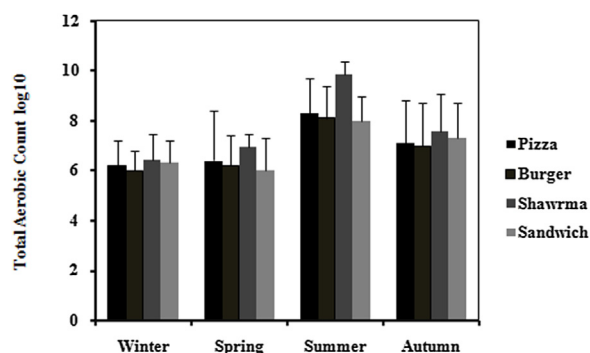


Figure 1. Total aerobic count of ready-to-eat food items during different seasons of the year.

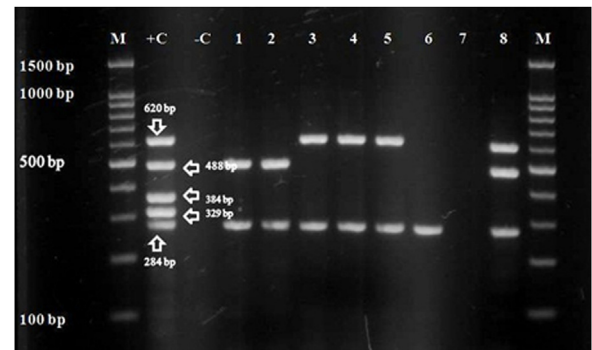


Figure 2. Multiplex polymerase chain reaction assay demonstrating the amplification of species-specific genes. The *Salmonella* *invA* gene conserved in all species is shown at 284 bp, with *S. paratyphi* A (flicA) at 329 bp, *S. paratyphi* B (SPAB) at 384 bp, *S. enteritidis* (sefA) at 488 bp and *S. typhimurium* (fliC) at 620 bp.

Discussion

Food consumed by humans should ideally be free of pathogens. Although RTE food sold by street vendors is cheap in price, its microbial quality may be compromised if proper hygienic measures are not adopted. In developing countries such as Pakistan, such measures are often neglected, and food items have been reported to be contaminated with several pathogens (Samad et al., 2018a, b, 2019, 2020). The results of the present study are alarming, as none of the RTE food samples tested ($n = 320$) were free of bacteria. The main reason for the high level of microbial contamination is likely to be poor sanitary conditions, as street vendors do not process or prepare the food items in proper shops. Food carts are often placed above drains, which serve as a potential source of microbial contamination. As food carts are on the streets, next to roads, food items for sale may be contaminated with dust and dung.

RTE food samples were found to be contaminated in every season of the year, but the problem was worse during the summer (Table 1). This is likely to be because foodborne pathogens are more prevalent during warm weather (Caponigro et al., 2010; Xylia et al., 2019). During summer, the average temperature in Quetta is 37°C . The abundance of house flies in summer could be another reason for high microbial contamination. The presence of house flies on both raw and prepared food items was observed during sample collection. Moreover, the personal hygiene of vendors, particularly sweating, during the summer further aggravates the situation. The association between high temperature and salmonellosis has been reported previously (Aik et al., 2018; Wang et al., 2018).

Of the RTE food samples analysed in this study, shawarma was found to be the most contaminated. A possible explanation for the high microbial content in this food item could be the cooking temperature and the raw materials used in the observed settings. Shawarma was mainly seen to be prepared without following the high holding temperature rule (Easa, 2010). In the observed scenarios, it was cooked on an ordinary hot plate, and the chicken, bread and sauces were not treated in accordance with recommended food safety procedures. Other factors could be the salad and the paper used to wrap the shawarma. In the case of sandwiches, it was observed that several vendors were selling pre-cooked products, and the buyer had no idea of the date or time of preparation as these details were not given in the observed settings. Food items that are pre-prepared and not stored properly are at greater risk of microbial contamination (Odu and Akano, 2012). Another observation which may contribute towards contamination is poor hand hygiene of vendors, many of whom

Table 2Antibiotic sensitivity of *Salmonella* spp. isolates in ready-to-eat food samples.

Antibiotics tested	<i>Salmonella</i> spp.					
	<i>Salmonella enteritidis</i> (n = 34)			<i>Salmonella typhimurium</i> (n = 15)		
	S	I	R	S	I	R
Gentamycin (10 µg)	21 (61.7%)	09 (26.4%)	04 (11.7%)	12 (80%)	03 (20%)	0 (0%)
Streptomycin (25 µg)	18 (52.9%)	06 (17.6%)	10 (29.4%)	01 (6.6%)	02 (13.3%)	12 (80%)
Kanamycin (30 µg)	25 (73.5%)	09 (26.4%)	01 (2.9%)	14 (93.3%)	01 (6.6%)	0 (0%)
Amoxicillin (25 µg)	0 (0%)	2 (5.8%)	32 (94.1%)	0 (0%)	01 (6.6%)	14
Cefotaxime (30 µg)	34 (100%)	0 (0%)	0 (0%)	15 (100%)	0 (0%)	0 (0%)
Erythromycin (15 µg)	0 (0%)	0 (0%)	34 (100%)	0 (0%)	0 (0%)	15 (100%)
Amoxicillin (30 µg)	16 (47%)	09 (26.4%)	10 (29.41%)	06 (40%)	05 (33.3%)	04 (26.6%)
Chloramphenicol (30 µg)	0 (0%)	0 (0%)	34 (100%)	0 (0%)	0 (0%)	15 (100%)
Oxytetracycline (30 µg)	18 (25.9%)	07 (20.5%)	09 (26.4%)	08 53.3%)	06 (40%)	01 (6.6%)
Ciprofloxacin (10 µg)	34 (100%)	0 (0%)	0 (0%)	15 (100%)	0 (0%)	0 (0%)

S, sensitive; I, intermediate; R, resistant.

do not wash their hands despite handling raw food, collecting money and packing cooked food.

The contamination of pizza was the most surprising as this is cooked in a hot oven at a high temperature. Contamination may be due to keeping the pizza at a lower temperature after cooking, sauce used after preparation, exposure to the dirty environment after preparation and before packing, and exposure to dirty packing material (Eslami et al., 2017).

Salmonellosis is one of the most prevalent diseases caused by ingestion of food items. This study evaluated the contamination of RTE food items with *Salmonella* spp. (Figure 2). As expected, the quantification of multiplex PCR demonstrated that *S. enteritidis* and *S. typhimurium* were the most prevalent serotypes in contaminated food items, which supports the findings of other studies (Isara et al., 2010; Morshdy et al., 2018; Yang et al., 2016).

Contamination of food items with bacterial pathogens is alarming, but the situation is worse when food items are contaminated with antibiotic-resistant strains. The *Salmonella* spp. isolated from food items in this study were sensitive to most of the antibiotics tested, but were resistant to certain groups of antimicrobials. Currently, the antibiotic sensitivity pattern of prevalent strains is not too alarming, and the results are in agreement with previous studies (Igbinosa et al., 2019; Onifade and Omololu, 2020).

In conclusion, the data suggest that RTE food items sold in Quetta are highly contaminated, and a significant number of samples were contaminated with pathogenic serotypes of *Salmonella* spp. There is a need to train street food vendors regarding food safety and quality in order to improve the safety of street food. The Government of Pakistan should take appropriate regulatory actions to improve the operating conditions of street food carts by thorough monitoring and the provision of facilities such as clean protected structures, potable clean water, and efficient waste collection and disposal systems.

Conflict of interest

None declared.

Funding

None.

Ethical approval

Not required.

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